

JAVEN LAI LE YU

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ABOUT ME

Analytical and detail-oriented individual with expertise in data analytics, machine learning, and database management. Passionate about leveraging technology to drive strategic decision-making and optimize efficiency by utilizing Gen-AI and automation.

EDUCATION

2022 – 2025

Diploma with merit in Big Data & Analytics, Temasek Polytechnic
GPA: 3.95 out of 4 (3rd in cohort)
CCA Grade: Distinction

2018 – 2021

GCE “O” Level Certificate, Tampines Secondary School
Obtained 7 credits.

SKILLS

Technical Skills

- **Data Analytics:** Develop **interactive dashboards** to uncover insights and build a **storyboard** to present findings effectively using **Tableau**.
- **Machine Learning:** Train and tune **predictive** and **classification** models like Random Forest, Ensemble, and Convolutional Neural Network (CNN) and interpret black-box predictive models using techniques like **SHAP**. Able to uncover insights from unsupervised learning methods like **Clustering, Rule Mining, and Topic Modelling**.
- **Databases:** Design **operational databases** and **data warehouse** architectures, and proficient in **SQL** for data extraction and management.
- **Automation:** Leverage technologies like **macro** and Robotic Process Automation (**RPA**) to perform repetitive tasks.
- **Web Development:** Built a **MERN Stack** web application from scratch. Implemented a text + image generating **ChatGPT wrapper** on an existing **Laravel** web application.

Generic Skills

- **Tech Savvy:** Well-versed in Windows OS and Microsoft 365 tools like PowerPoint, Excel, Word, and Outlook.
- **Team Player:** Collaborated with diverse teams on school, hackathon, and workplace projects, demonstrating strong interpersonal and teamwork abilities.
- **Adaptability & Initiative:** Actively participated in hackathons, took on part-time jobs, and explored new technologies beyond the curriculum.
- **Problem-Solving & Critical Thinking:** Skilled in breaking down complex issues logically and leveraging AI tools to solve problems and improve efficiency.
- **Fast Learner:** Able to grasp concepts and learn to use new software quickly.

WORK EXPERIENCE

Mar 2025 – May 2025	BoardRoom IT Assistant (Contract) • Set up IT equipment and software applications for meetings. • Troubleshoot IT issues during meeting registration.
Dec 2024 – Jan 2025	Ilao Ilao Service Crew (Part-Time) • Trained to service customers professionally. • Operated the cashiering system to process transactions seamlessly. • Assisted in store operations like preparation, cleaning, restocking, and serving.
Apr 2024 – Oct 2024	Synapse RPA Developer (Intern) • Spearheaded technical implementation of UiPath automation projects at IMH. • Developed and maintained IT solutions based on department requirements. • Involved in the whole project lifecycle from defining project requirements with users to performing User Acceptance Testing (UAT).
Mar 2023 - Apr 2023	UPS Supply Chain Solutions Warehouse Assistant (Part-Time) • Processed transaction invoices to generate Excel reports. • Performed Data Entry using SAP HANA.

CERTIFICATIONS

Mar 2025	Fundamentals of Deep Learning by NVIDIA
Jul 2024	Food Hygiene Certification (L1)
Oct 2023	Google Analytics for Beginners
Sep 2023	Advanced Proficiency in KNIME Analytics Platform (L2)
Sep 2023	Alteryx Designer Core Certification
Nov 2022	NDG Linux Unhatched

AWARDS/ACHIEVEMENTS

2024	Gold Award for SUSS Analytics and Visualisation Competition 2024 <i>Built an interactive Tableau Dashboard to uncover insights and proposed strategies to help the business improve sales performance.</i>
2023	Bronze Medal for Poly-ITE Olympiad for Informatics 2023 <i>Solved data structures and algorithms problem statements by writing efficient code.</i>
2023	Winner for Sustainability Campaign Challenge <i>Collaborated with a teammate to design and pitch a campaign to reduce the usage of plastic water bottles within Temasek Polytechnic.</i>
2017	Edusave EAGLES Award <i>Demonstrated leadership qualities, service to community and schools, excellence in non-academic activities, and good conduct.</i>
2017	SYF Certification of Recognition – Art Exhibition

CO-CURRICULAR ACTIVITIES

2022 – 2023	IT Software Solutions for Business WorldSkills SG Trainee Trained in ASP.NET (C#) and MSSQL to develop an Inventory Management System.
2018 - 2021	MakerFaire Sub-Committee Member Participated in workshops and hackathons, gaining hands-on experience in programming, drone operation, and 3D printing.

PROJECTS

Data Visualization and Analytics	Dashboard to analyze sales trends and patterns of a Car Service Company (Gold Award): https://github.com/Javen05/SUSS-AVC2024 [Interactive Dashboard] Dashboard and Storyboard about OCBC performance: https://github.com/Javen05/Data-Storytelling [Interactive Dashboard, Storyboard] Dashboard about HDB resale prices in Central Area (Bronze Award): https://github.com/Javen05/SUSS-AVC2023 [Interactive Dashboard] Created a Dashboard to support the problem statement: https://github.com/Javen05/sustainabilityCampaign [Dashboard]
Machine Learning (ML)	Conducted Text Analytics methods like Sentiment Analytics, Named Entity Extraction, Text Clustering, Topic Discovery, and Text Classification on data scraped from Reddit's r/artificial to determine if people are ready for the era of AI: https://github.com/Javen05/TSAL [Text Analytics] Developed an Image Classification CNN with 92% overall accuracy to distinguish between benign and malignant moles: https://www.youtube.com/watch?v=ww2qCIFy5N8 [CNN, Android Dev] Developed (1) a Gen-AI-powered questionnaire generator and (2) a question analytics algorithm that produces qualitative insights and actionable recommendations for my Polytechnic Final Year Project. [Gen-AI, Laravel] Performed supervised and unsupervised ML to derive strategies to improve the operations of a bicycle rental company: https://github.com/Javen05/Data-Science [Predictive, Clustering] Engineered a Random Forest Regressor with business-related features and interpreted the model using SHAP to uncover insights on how factors influence Q4 sales of a retail hypermarket: https://github.com/Javen05/Machine-Learning [Predictive, SHAP] Conducted analysis on Corporate Sustainability by building and interpreting supervised ML models on SAS Enterprise Miner: https://github.com/Javen05/Corporate-Sustainability-Analysis [Predictive, Clustering, Business Analytics, Data Mining] Pitched an ML model to predict upcoming cargo traffic: https://github.com/Javen05/PSA-CODE-SPRINT-2023 [Predictive]

Data Pipeline + Data Warehouse	Used Alteryx to perform ETL and store data in a locally hosted MySQL server. Extracted data from Data Warehouse to conduct analysis using Alteryx to provide business insights: https://github.com/Javen05/DWBI [Data Warehouse, Data Visualization, Business Intelligence] Built an automated Data Pipeline using AWS that stores data in a star schema and hosts a prepared dataset as a public REST API: https://github.com/Javen05/AWS-Experience [AWS, Data Warehouse, Data Pipeline, Dashboard]
Web and App Development	Full Stack restaurant review website using MySQL, Express, React, and NodeJS: https://github.com/Javen05/Foodie [Frontend Dev, Backend Dev, Designing Operational Database] Designed a music player app concept using Adobe XD and tried building it on Android Studio using Java: https://github.com/Javen05/MYSIC [Adobe XD Wireframe, Android Dev] Pitched an Adobe XD app prototype for laypersons to get legal knowledge: https://github.com/Javen05/legalsg [Adobe XD Wireframe]